Kiwi bank. Business Banking.

Kiwibank Direct Credit & Direct Debit File Format Guide

A guide to building a file compatible with Kiwibank Internet Banking for Business (IB4B)

Feb 2021

Contents

Purpose. 3
Overview 3
Glossary of Terms 4
Direct Credits - Overview 5
Key Requirements 5
Direct Credits File Format 6
Header Record – Direct Credits 6
Transaction Record – Direct Credits 7
Transaction Record – Account Number Format – Direct Credits 8
Trailer Record – Hash Total – Direct Credits 9
Trailer Record – Direct Credits 10
Direct Debits Overview 11
Key Requirements 11
Direct Debits File Format 12
Header Record – Direct Debits 12
Transaction Record – Direct Debits 13
Transaction Record – Account Number Format – Direct Debits 14
Trailer Record – Hash Total – Direct Debits 15
Trailer Record – Direct Debits 16
Modulus Check Specifications 17
C1. Validation of codelines 17
C2. Codeline Check Digit Routines 17
Modulus 11 check digit 17
Format 11A 17
Format 11B 18
Format 11C 19
Format 11D 19
Modulus 10 check digit 21
Format 10A 21
Disclaimer 22

Purpose.

This document details the file format required by Kiwibank to process debit or credit batches, including Payroll.

Overview

The files consist of the following sections:

- Header (type 1 record)
- Transaction Record (type 2 record)
- Trailer Record (type 3 record)

The file will contain the Header and Trailer and at least one Transactional record. The transactional records may be either debit or credit, denoted by the Tran Code field. The field codes can be found in Glossary of Terms.

The file can contain exclusively Direct Credit or Direct Debit records, it cannot contain both in the same file.

Files processed in this manner cannot be edited once they have been submitted.

Each section consists of comma delimited ASCII information fields. The length of each field being bound by a maximum field length (refer to tables detailed in each section).

Please contact the Kiwibank Business Direct team on 0800 601 601, or your Relationship Manager for assistance with validating the file format.

Glossary of Terms

This table describes some of the terms used throughout this document.

Term	Description
A	Alphanumeric: this field can include any of the following letters: a - z and A - Z Numbers: 0 - 9 Special characters: (+ $ * !) ; - / ? : ' = & < > . _ @ # % SPACE
Codelines	Another name for account number.
CRLF	A carriage return, line feed character (CRLF) must follow each record (HEX: 0D 0A).
DC	Direct Credit. This includes Payroll transactions.
DD	Direct Debit.
File Size	Each file is limited to 10MB of data. This equates to between 10,000 and 12,000 lines in a batch. The overall file size is the determining factor.
Filename	The preferred file name will have a ".kbb" file extension. However, ".txt" will also work if the file is correctly formatted.
Hash Total	The sum of the payment account numbers in the file.
Header	The first line in the batch that is identified by the record type 1. This contains the account to be debited or credited, the type of transactions and the date information.
IB4B	Kiwibank Internet Banking for Business.
N	Numeric: this field can only contain numbers 0 – 9.
Payee	Party receiving the credit of funds.
Payer	Party being debited funds.
Trailer	The last line of data in the batch that is identified by the record type 3. This contains the transaction count, value of the batch and the hash total. There must be one line after this record that is blank.
Tran Code	The Payments New Zealand Transaction code for the type of transaction. The common codes are <u>Credits</u> 50 Direct Credit. 52 Payroll Credit. <u>Debits</u> 00 Direct Debit.
Transaction	Each payment record within the file which is identified by the record type 2.

Direct Credits - Overview

The Direct Credit import file facility allows you to transfer data from your organisation's accounting system directly into IB4B for processing.

Direct credit files can be used to pay any New Zealand domestic bank account from your Kiwibank transactional account.

Key Requirements

Direct credit files must adhere to the following standards to be successfully imported to IB4B:

- The files are comma delimited ASCII files
- Each file must contain a Header record, one or more Transaction records and a Trailer record
- Each record must terminate with a carriage return line feed character (CRLF)
- Embedded spaces are permitted within each field but not trailing spaces
- As each field is separated by commas, therefore commas are not permitted within a field.
- The file extension must be either .kbb or .txt
- The maximum size of the batch is limited to **10MB** of data. This equates to between 10,000 and 12,000 lines (Transaction records) in the batch
- The following characters are permitted in the filename:

  a - z, A - Z, 0 - 9, space - ' & , # _ .
- Transaction amounts must be greater than zero
- The file format is the same for creditor payments as well as Payroll

The header record contains information about the account to be debited, the type of transactions (Direct Credits).

Example 1,,,,389016070450500,7,210216,210421,

Field Type	Data Type	Max Record Length	Description	Comment	Default
Record Type	Numeric	1 character	Type of record	1 denotes the header	1
Auth Code	Numeric	7 characters	Blank for DC Batches	Blank for DC Batches	No default
Batch Number	Numeric	2 characters	Customer Selected batch Number	Optional	No default
Batch Sequence Number	Numeric	4 characters	Customer Selected sequence Number	Optional	No default
Account Number	Numeric	16 characters	Account number that is to be debited to for DC batch	Kiwibank customer's account number.	No default
Batch Type	Numeric	1 character	Batch type	Always "7" for DC batches	No default
Batch due date	Numeric	6 characters	YYMMDD	Valid date (not a weekend or public holiday) and not in the past. Can be up to 65 days in advance.	No default
File date	Numeric	6 characters	YYMMDD	Valid date (not a weekend or public holiday) and not in the future	No default
Indicator	Alphanumeric	12 characters	Reporting indicator	Not used	Left Blank

Transaction Record – Direct Credits

The transaction record contains information about the other party (beneficiary) to be credited, and the information to be shared with the other party.

Example 2,389016072577900,50,12,Demo 2 KB Transactio,Test -,,,Test,Demo Company,Test,,Test

Field Type	Data Type	Max Record Length	Description	Comment	Default
Record Type	Numeric	1 Character	Type of Record	2 denotes the transaction record	2
Account Number	Numeric	16 Characters	Payee account number DC	Must pass the modulus 11 check used by New Zealand banks. For valid format refer to Transaction Record – Account Number Format section. For Modulus 11 check specifications, please refer to the Modulus 11 check digit section.	None
Tran Code	Numeric	2 Characters	Payments New Zealand transaction code 50 for Direct Credits. 52 for Payroll Credits	Refer to the description for the Transaction codes	None
Amount	Numeric	12 Characters	Transaction amount in cents	No decimal point	None
Other Party Name	Alphanumeric	20 Characters	Payee name for DC		None
Other Party Reference	Alphanumeric	12 Characters	Reference Details	Optional	None
Other Party Code	Alphanumeric	12 Characters	Code Details	Optional	None
Other Party Alpha Ref			Blank	Blank	Blank
Other Party Particulars	Alphanumeric	12 Characters	Particulars Details	Optional	None
This Party Name	Alphanumeric	20 Characters	Name of Payer for DC		None
This Party Code	Alphanumeric	12 Characters	This Party Code Details	Optional	None
This Party Reference	Alphanumeric	12 Characters	This Party Reference Details	Optional	None
This Party Particulars	Alphanumeric	12 Characters	This Party Particulars Details	Optional	None

There are two different options when it comes to formatting account numbers in payments (Direct Credit or Direct Debits). Option One has two digits in the suffix whilst Option Two has three digits in the suffix. The banking industry in New Zealand has agreed interchange two digits for the suffix going forward and therefore Option One is the preferred option.

For this guide, the following key is offered to assist with correctly formatting account numbers.

B = Bank Number	This will always be two numbers
b = Branch Number	This will always be four numbers
a = Account number	This can be either six or seven numbers. If there are only six numbers, add a leading zero at the start of the account number **space**. **Note:** the first position should therefore always be a zero
s = Suffix	This will be either two numbers of three numbers. If there are three digits in the suffix, the first suffix position should always contain a “0”

	OPTION ONE	OPTION TWO
Format	**BBbbbbaaaaaaass**	**BBbbbbaaaaaaasss**
Example	**389016072577900**	**3890160725779000**

Trailer Record – Hash Total – Direct Credits

The hash total is made up of the sum of those digits in positions 3 to 13 inclusive of the account number from each transaction. The first two and last two or three digits of each account number are ignored. It is eleven characters in length.

If the number is larger than eleven digits, drop the most significant numbers i.e. those on the left. If the number is less than eleven digits then zero fill to eleven, the zeroes being left aligned.

For example:

38 **90160725779** 000

38 **90160704505** 000

The above account numbers generate a hash total of **80321430284**

Trailer Record – Direct Credits

The trailer record, sometimes called a control record, contains information that validates the data within the batch.

Example 3,50145116,10014,69507950906

Field Type	Data Type	Max Record Length	Description	Comment	Default
Record Type	Numeric	1 Character	Type of Record	3 denotes the trailer record	3
Transaction Amount	Numeric	20 Characters	Total of Type 2 records in cents	No Decimal Point	None
Transaction Count	Numeric	6 characters	Count of Type 2 Records		None
Hash Total	Numeric	11 Characters	Hash total of Type 2 Records	Refer to the note below	None

Direct Debits Overview

The Direct Debit import file facility allows you to upload data from your organisation's accounting system directly into IB4B for processing.

Direct Debit files can be used to credit your nominated Kiwibank domestic bank account.

Direct Debit payment files can debit funds from any New Zealand domestic bank account, that you have the account owner's permission to debit.

Note before you can upload a direct debit file for processing you must have access to Kiwibank Direct Debit Services.

Key Requirements

Direct debit files must adhere to the following standards to be successfully imported by IB4B:

- The files are comma delimited ASCII file
- Each file must contain a Header record, one or more Transaction records and a Trailer record
- Each record must terminate with a carriage return line feed character (CRLF)
- Embedded spaces are permitted within each field but not as trailing spaces
- As each field is separated by commas, commas are not permitted within a field.
- The file extension must be either .kbb or .txt
- The maximum size of the batch is limited to 10MB of data. This equates to between 10,000 and 12,000 lines (Transaction records) in the batch
- The following characters are permitted in the filename:

  a - z, A - Z, 0 - 9, space - ' & , # _ .
- Transaction amounts must be greater than zero
- The file format is the same for creditor payments with the exception of a changed batch type and the inclusion of a direct debit authorisation code (Auth Code)

Direct Debits File Format

Header Record – Direct Debits

The header record contains information about the account to be debited, the type of transactions (Direct Credits).

Example 1,3899969,,,389016070450500,6,200717,200717,

Field Type	Data Type	Max Record Length	Description	Comment	Default
Record Type	Numeric	1 character	Type of record	1 denotes the header	1
Auth Code	Numeric	7 characters	DD Auth Code for all DD Batches	DD Auth Code required for DD batches	No default
Batch Number	Numeric	2 characters	Customer Selected batch Number	Optional	No default
Batch Sequence Number	Numeric	4 characters	Customer Selected sequence Number	Optional	No default
Account Number	Numeric	16 characters	Account number that the batch will be credited to for DD	Kiwibank customer's account number.	No default
Batch Type	Numeric	1 character	Batch type	Always "6" for DD batches	No default
Batch due date	Numeric	6 characters	YYMMDD	Valid date (not a weekend or public holiday) and not in the past. Can be up to 65 days in advance.	No default
File date	Numeric	6 characters	YYMMDD	Valid date (not a weekend or public holiday) and not in the future	No default
Indicator	Alphanumeric	12 characters	Reporting indicator	Not used	Left Blank

The transaction record contains information about the other party (payer/acceptor) to be debited, and the information to be shared with the other party.

Example 2,389016072577900,50,12,Demo 2 KB Transactio,Test -,,,Test,Demo Company,Test,,Test

Field Type	Data Type	Max Record Length	Description	Comment	Default
Record Type	Numeric	1 Character	Type of Record	2 denotes the transaction record	2
Account Number	Numeric	16 Characters	Payer account number DD	Must pass the modulus 11 check used by New Zealand banks. For valid format refer to Transaction Record – Account Number Format section. For Modulus 11 check specifications, please refer to the Modulus 11 check digit section.	None
Tran Code	Numeric	2 Characters	Payments New Zealand transaction code 00 for Direct Debits	Refer to description for the Transaction codes	None
Amount	Numeric	12 Characters	Transaction amount in cents	No decimal point	None
Other Party Name	Alphanumeric	20 Characters	Payer name for DD		None
Other Party Reference	Alphanumeric	12 Characters	Reference Details	Optional	None
Other Party Code	Alphanumeric	12 Characters	Code Details	Optional	None
Other Party Alpha Ref			Blank	Blank	Blank
Other Party Particulars	Alphanumeric	12 Characters	Particulars Details	Optional	None
This Party Name	Alphanumeric	20 Characters	Name of Initiator for DD		None
This Party Code	Alphanumeric	12 Characters	This Party Code Details	Optional	None
This Party Reference	Alphanumeric	12 Characters	This Party Reference Details	Optional	None
This Party Particulars	Alphanumeric	12 Characters	This Party Particulars Details	Optional	None

There are two different options when it comes to formatting account numbers in payments (Direct Credit or Direct Debits). Option One has two digits in the suffix whilst Option Two has three digits in the suffix. The banking industry in New Zealand has agreed interchange two digits for the suffix going forward and therefore Option One is the preferred option.

For this guide, the following key is offered to assist with correctly formatting account numbers.

B = Bank Number	This will always be two numbers
b = Branch Number	This will always be four numbers
a = Account number	This can be either six or seven numbers. If there are only six numbers, add a leading zero at the start of the account number **space**. **Note:** the first position should therefore always be a zero
s = Suffix	This will be either two numbers of three numbers. If there are three digits in the suffix, the first suffix position should always contain a “0”

	OPTION ONE	OPTION TWO
Format	**BBbbbbaaaaaaass**	**BBbbbbaaaaaaasss**
Example	**389016072577900**	**3890160725779000**

The hash total is made up of the sum of those digits in positions 3 to 13 inclusive of the account number from each transaction. The first two and last two or three digits of each account number are ignored. It is eleven characters in length.

If the number is larger than eleven digits, drop the most significant numbers i.e. those on the left. If the number is less than eleven digits then zero fill to eleven, the zeroes being left aligned.

For example:

38 **90160725779** 000

38 **90160704505** 000

The above account numbers generate a hash total of **80321430284**

Trailer Record – Direct Debits

The trailer record, sometimes called a control record, contains information that validates the data within the batch.

Example 3,50145116,10014,69507950906

Field Type	Data Type	Max Record Length	Description	Comment	Default
Record Type	Numeric	1 Character	Type of Record	3 denotes the trailer record	3
Transaction Amount	Numeric	20 Characters	Total of Type 2 records in cents	No Decimal Point	None
Transaction Count	Numeric	6 characters	Count of Type 2 Records		None
Hash Total	Numeric	11 Characters	Hash total of Type 2 Records	Refer to the note below	None

C1. Validation of codelines

To enable validation of codelines and Settlement of paper, the validation routines of Settlement Members and Appointors must be advised to all other Settlement Members through the Association.

C2. Codeline Check Digit Routines

The current check digit validation routines approved by the Association for use by Settlement Members or their Appointors are set out below.

Modulus 11 check digit

The following four check digit validation routines use a Modulus 11 check digit.

Format 11A

(i) This format is used by bank numbers 01, 02, 03, 06, 09 (where branch number is not 0000), 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 27, 30, 31, 34, 36 and 38

(ii) The account structure is:

2 digit bank 4 digit branch 6 digit account 2 digit suffix

(iii) The account check digits are performed over the four digits of the branch number and six digits of the account number

(iv) The weightings for the calculation are from left to right: 6,

3, 7, 9, 10, 5, 8, 4, 2, 1

(v) The products are added together and the sum divided by 11. If the remainder is zero the account check digits, e.g. Account Number 01-0902-0068389-00

Codeline Position	Digit	Weighting	Product
34	0	6	0
33	9	3	27
32	0	7	0
31	2	9	18
27	0	10	0
26	6	5	30
25	8	8	64
24	3	4	12
23	8	2	16
22	9	1	9
			176

(vi) Dividing by 11 gives 16 with a remainder of 0 (no remainder). Thus, the check digit is correct.

Format 11B

(i) This format is used by bank 08

(ii) The account structure is:

2 digit bank, 4 digit branch, 7 digit account, 3 digit suffix

(iii) The account check digits are performed over the seven digits of account number only

(iv) The weightings for the calculations are from left to right:
7, 6, 5, 4, 3, 2, 1

(v) The products are added together and the sum divided by 11. If the remainder is zero the account check digits, e.g.
Account Number 08-6523-1954512-001

Codeline Position	Digit	Weighting	Product
28	1	7	7
27	9	6	54
26	5	5	25
25	4	4	16
24	5	3	15
23	1	2	2
22	2	1	2
			121

(i) Dividing by 11 gives 11 with a remainder of 0 (no remainder). Thus, the check digit is correct

Format 11C

(i) This format is used by bank numbers, 01, 02, 03, 06, 09 (where branch number is not 0000), 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 27, 30, 31, 34, 36 and 38 but only for accounts that fall within the range 990000 - 999999.

(ii) The account structure is:

2 digit bank 4 digit branch 6 digit account 2 digit suffix

(iii) The account check digits are performed over the six digits of account number only

(iv) The weightings for the calculation are from left to right: 10, 5, 8, 4, 2, 1

(v) The products are added together and the sum divided by 11. If the remainder is zero the account check digits, e.g. Account Number 01-0902-0998907-00

Codeline Position	Digit	Weighting	Product
27	9	10	90
26	9	5	45
25	8	8	64
24	9	4	36
23	0	2	0
22	7	1	7
			242

(vi) Dividing by 11 gives 22 with a remainder of 0 (no remainder). Thus the check digit is correct

Format 11D

(i) This format is reserved for use by bank 09 where the branch number is 0000

(ii) The account structure is:

2 digit bank 4 digit branch 4 digit account 1 digit suffix

(iii) The account check digits are performed over the four digits of the account and one digit of the suffix

(iv) The weightings for the calculation are from left to right: 5, 4, 3, 2, 1

(v) The digits of each product are added together. This step is repeated for any result that has more than one digit. The results of these additions are added and the sum is divided by 11. If the remainder is zero the account check digits, e.g. Account Number: 09-0000-00007471-2

Codeline Position	Digit	Weighting	Product	Sum
25	7	5	35	8
24	4	4	16	7
23	7	3	21	3
22	1	2	2	2
20	2	1	2	2
				22

(vi) Dividing by 11 gives 2 with a remainder of 0 (no remainder). Thus the check digit is correct

(vii) In addition to checking the account number it also necessary to check the serial number

(viii) Two formats of serial numbers are used, one that requires check digit verification and one that does not. If the serial number is other than one of the below it is invalid

(ix) Valid, No Check Digit

Any 6 digit number, codeline positions 44 - 39

(x) Valid, Requires Check Digit

Any 8 digit number, codeline positions 42 - 39, e.g 7 digit serial number plus a check digit

(xi) The serial number check digit is calculated by dividing the 7 digits by 9 and subtracting the remainder from 9. The result is then the check digit, e.g.

Serial Number: 5278034-7

5278034/9 = 586448 remainder 2

9 - 2 = 7

(xii) Remainder of 7 equals check digit therefore serial number check digits

Modulus 10 check digit

The following two check digit validation routines use a Modulus 10 check digit.

Format 10A

(i) This format is used by bank 25

(ii) The account structure is:

2 digit bank 4 digit branch 7 digit account 3 digit suffix

(iii) The account check digits are performed over the account number only

(iv) The weightings for the calculation are from left to right: 1, 7, 3, 1, 7, 3, 1

(v) The products are added together and the sum divided by 10. If the remainder is zero the account check digits, e.g.
Account Number: 25-2500-2608486-001

Codeline Position	Digit	Weighting	Product
28	2	1	2
27	6	7	42
26	0	3	0
25	8	1	8
24	4	7	28
23	8	3	24
22	6	1	6
			110

(vi) Dividing by 10 gives 11 with a remainder of 0 (no remainder). Thus the check digit is correct

Disclaimer

This publication has been provided for general information only.

Although every effort has been made to ensure this publication is accurate, the contents should not be relied upon or used as a basis for entering into any products or processes described in this publication. Kiwibank strongly recommends readers seek independent legal/financial/technical advice prior to acting in relation to any of the matters discussed in this publication.

Neither Kiwibank nor any person involved in this publication accepts any liability for any loss or damage whatsoever which may directly or indirectly result from any advice, opinion, information, representation or omission, whether negligent or otherwise, contained in this publication.